



REVIEW ARTICLE

Amaranthaceae microgreens: A sustainable source of phytonutrients and bioactive compounds

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Abstract

Microgreens, often referred to as "vegetable confetti," are early-stage edible seedlings derived from a wide range of vegetables, grains and herbs. In recent years, they have gained considerable attention due to their remarkable nutritional value and rich bioactive profile. This review focuses on Amaranthaceae microgreens, highlighting their cultivation techniques, nutritional composition, bioactive constituents and associated health benefits. Amaranthaceae microgreens, including species such as *Amaranthus cruentus* L., *Chenopodium quinoa* Willd., *Beta vulgaris* L. and *Spinacia oleracea* L., are rich in essential vitamins and minerals such as calcium, magnesium, potassium and iron. They also contain diverse bioactive compounds, including phenolics, flavonoids, carotenoids, betalains and anthocyanins, many of which exhibit strong antioxidant activity. These compounds contribute to multiple health-promoting effects, including antioxidant, anti-inflammatory, antimicrobial and anticancer activities. Cultivation methods, such as soil-based systems, hydroponics and controlled-environment agriculture, significantly influence the growth, yield and phytochemical composition. Optimisation of factors such as light spectrum, photoperiod and nutrient supply enhances biomass production and bioactive compound accumulation. Overall, Amaranthaceae microgreens show strong potential as sustainable functional foods for improving human health and addressing global food security challenges.

Keywords: amaranthaceae microgreens; bioactive compounds; cultivation techniques; functional foods; health benefits; nutritional composition; phytochemicals

Introduction

Microgreens are a specialised crop grown on a microscale, mainly obtained from the seeds of vegetables, grains and herbs (1). Depending on the crop type, their rapid growth period lasts 2–3 weeks, resulting in the formation of a central stem, two cotyledon leaves and often the first pair of true leaves (2). According to studies, the species used to produce microgreens belong to several different botanical families, including Amaranthaceae, Amaryllidaceae, Apiaceae, Asteraceae, Brassicaceae, Cucurbitaceae, Fabaceae and Lamiaceae (3–7). In terms of growth, microgreens generally require less light than fully grown mature plants, enabling the use of sustainable cultivation methods such as hydroponic systems, optimised lighting and controlled environmental conditions, which can lead to regulation of phytonutrient accumulation and reduction of anti-nutrient content in microgreens (8). Microgreens can be cultivated year-round under controlled conditions, as their growth is primarily influenced by factors such as light, temperature and nutrient availability rather than seasonal variation (9, 10). Initially, microgreens were used as nutrient-rich greens to enhance the visual appeal of dishes and can be consumed in various forms (11). Currently, microgreens are valued for their high nutrient content, particularly phenolic compounds, carotenoids and vitamin C, E and K1, as well as macro- and micro-nutrients (8). Comparative studies have shown that microgreens often surpass their mature

counterparts in nutritional value (11). For example, microgreens from two lettuce cultivars have demonstrated higher levels of calcium, magnesium and total polyphenols compared to mature plants. Similarly, other studies have reported elevated levels of macro- and micro-nutrients as well as antioxidant compounds in vegetable microgreens (8).

Studies on Amaranthaceae microgreens have been limited to certain species, although the family comprises over 165 genera and more than 2000 species, distributed worldwide in tropical and temperate regions (12). This family, belonging to the order Caryophyllales, is one of the most morphologically diverse plant families (13). Members are predominantly herbaceous, although shrubs, small trees and vines are also present, with simple leaves and small clustered flowers arranged in diverse inflorescences (14). Representative species commonly used for microgreen production, such as *Amaranthus cruentus* L., *Beta vulgaris* L., *Chenopodium quinoa* Willd. and *Spinacia oleracea* L. belong to this family (Fig. 1). Many species have been traditionally used as medicinal herbs across temperate, tropical and subtropical regions (15). Extracts from Amaranthaceae plants have demonstrated antioxidant, antidiabetic, immunostimulatory, anticancer, antimicrobial, analgesic, anti-inflammatory, hypolipidemic, diuretic, antihypertensive and hypoglycemic properties (16–24). These plants contain bioactive compounds such as diterpenes, volatile oils, phenols, betacyanins



Fig. 1. Representative species of the Amaranthaceae family commonly used for microgreen production.

and flavonoids (15, 25, 26). Well-known genera include *Amaranthus*, *Bassia*, *Beta*, *Gomphrena*, *Chenopodium*, *Salicornia*, *Celosia*, *Atriplex*, and *Spinacia* (26, 27). The plants from this family are important due to their high nutritional value, rich bioactive composition and their potential role in improving human health and food security. Additionally, they possess considerable economic importance, as leafy vegetables, pseudocereals, ornamental plants, and traditional medicinal resources (13).

Research on Amaranthaceae microgreens has primarily focused on a few species, red amaranth, spinach, quinoa, red beet and Swiss chard. Based on previous work carried out by various researchers, this review synthesises current research on the cultivation methods, nutritional profiles and bioactive compounds of Amaranthaceae microgreens, highlighting their relevance in health promotion and sustainable food systems.

Cultivation techniques for Amaranthaceae microgreens

Several cultivation techniques have been developed to enhance the yield and nutritional quality of microgreens, including traditional soil-based systems, hydroponic systems, controlled environment agriculture and aeroponic systems (28). The growth of microgreens

is influenced by environmental factors such as temperature, humidity, light and moisture availability, with optimal development achieved under controlled conditions that support germination, growth and phytochemical accumulation (29–31). In controlled cultivation systems, rainfall is not a direct factor, however, adequate and uniform moisture supply through irrigation is essential for successful germination and development (10). Each cultivation method affects the growth rate, nutritional composition and phytochemical accumulation differently (Table 1).

A conventional soil-based method uses fertilisation to enhance of macro- and micro-nutrient content of crops. Several studies have investigated this method of cultivation in the production of *A. cruentus* (red amaranth) and *B. vulgaris* (beet) microgreens. For red amaranth microgreens, the use of different soil types showed that black soil reduced the harvest time to 8 days and produced 2.6 times greater stem length and fresh weight (FW) compared to orange and yellow soils with coconut compost under various light sources (32). Similarly, the use of cocopeat-organic manure mixture and sterilisation of seeds with 2 % sodium hypochlorite for beet microgreens showed the most nutritional parameters peaked at the 15th day (33).

Table 1. Cultivation techniques and their effects on Amaranthaceae microgreens

Microgreens	Cultivation method	Conditions	Effects on microgreens	References
<i>A. cruentus</i> (Red amaranth)	Soil-based with coconut compost	Black, orange and yellow soils under varied lighting (sunlight, red + blue and white artificial light)	Black soil produced 2.60 × higher stem length and 2.13 × higher fresh weight	(32)
	Soil-based indoor cultivation	Cocopeat with organic manure mix, sterilised seeds, 15-day harvest	Peak nutrient content observed on day 15, improved chlorophyll, polyphenol and antioxidant levels	(33)
<i>B. vulgaris</i> (Beet)	Hydroponics	Vermiculite, peat-lite and fertilisation with calcium nitrate	Increased germination rate and yield (10.14 kg/m ²)	(34)
	Controlled environment	LED spectrum (75 % red, 23 % blue, 2 % far-red), photoperiod (12–16 hr), varying light intensities	Optimised yield and improved energy and water efficiency	(35)
<i>B. vulgaris</i> L. subsp. <i>vulgaris</i> (Swiss chard)	Floating hydroponics	Hoagland's nutrient solution under polyethylene cover and 17-day harvest	Tender shoots with enhanced nutrient content	(36)
<i>Amaranthus</i> spp.	Controlled environment	Light spectrum (red, blue, red + blue, white), light intensity (130–280 μmol m ⁻² s ⁻¹), photoperiod (8–20 hr)	Increased fresh yield and improved vitamin C, anthocyanins and total carotenoids	(37)
<i>A. cruentus</i> 'copperhead'	Controlled environment	LED intensity and photoperiod adjustments (PPFD, DLI)	Increased biomass, leaf area, and nutrient efficiency	(38)

LED: light emitting diode; **PPFD:** photosynthetic photon flux density; **DLI:** daily light integral.

In hydroponic systems, water serves as the primary medium and substrates such as coconut fiber, perlite, vermiculite, rockwool are commonly used (28, 39, 40). In beet microgreens, hydroponic cultivation produced a yield of 10.14 kg/m² at 15th day after planting, demonstrating advantages over conventional methods (34). The floating hydroponic technique (deep water culture) has been used for Swiss chard microgreens, where germinated seeds in vermiculite are maintained at 24 °C under a polyethylene roof and supplied with half-strength Hoagland's nutrient solution, resulting in tender and nutrient-rich shoots at 17th day (36).

In controlled environment agriculture, the artificial lighting systems such as fluorescent tubes, high-intensity discharge lamps and horticultural light-emitting diode (LED) play a crucial role in regulating plant growth. Light spectrum, intensity and photoperiod significantly influence both growth and nutritional quality of microgreens (41). For example, red amaranth microgreens exposed to a red-to-blue light ratio of 70:30 at 280 µmol m⁻²s⁻¹ for 16 hr per day showed increase yield and enhanced levels of vitamin C, anthocyanins and carotenoids (37). Similarly, *A. cruentus* 'copperhead' microgreens showed variations in growth under different LED conditions. Shorter stems were observed under longer photoperiods, while biomass and leaf area were influenced by daily light integral (DLI) and photosynthetic photon flux density (PPFD) (38). In another study, beet microgreens cultivated under controlled LED lighting achieved the highest yield of 482.73 g/m² with a 12-hr photoperiod and a light spectrum consisting of 75 % red, 23 % blue and 2 % far-red. However, photoinhibition was observed at 220 µmol m⁻²s⁻¹, indicating the importance of precise light management for optimising growth and nutritional quality (35).

Nutritional and phytochemical composition

The relationship between diet and health is a widely recognised phenomenon. Epidemiological studies have found that the consumption of fruits and vegetables is associated with a reduced risk of chronic diseases such as diabetes, obesity and hypertension (42). Microgreens are considered a rich source of bioactive compounds with significant nutritional benefits (43). Their increasing demand is attributed to their distinct chemical composition compared to mature plants (44–46). Microgreens are rich in both macro- and micro-nutrients, including elements such as iron, zinc, potassium, calcium, nitrogen, phosphorus, sulfur, manganese, selenium and molybdenum (47–49). Furthermore, they contain a wide range of bioactive compounds, including ascorbic acid (50, 51), phyloquinones (52), α-tocopherol (50, 53), β-carotene (54), carotenoids (55, 56), phenolics (8, 57, 58), anthocyanins (5), glucosinolates (57, 58) and sugars (55), which contribute to their health-promoting properties. In addition to their rich nutritional composition, the stability and availability of these bioactive compounds can be influenced by post-harvest processing and cooking methods (59). Thermal treatments may reduce water-soluble vitamins, phenolics and antioxidant activity due to thermal degradation and leaching, while milder methods such as steaming or microwaving may enhance nutrient bioavailability by reducing anti-nutritional factors and enhancing digestibility (59, 60). The diverse nutritional and phytochemical content found in different types of Amaranthaceae microgreens is elaborated upon in this section (Table 2).

The proximate composition of quinoa (*Chenopodium quinoa*) microgreens shows higher levels of protein content (28.2–37.0 %) compared to grains (9.1–15.7 %) (61). Moderate levels of fiber (6.9–7.8 %), fats (2.4–4.5 %), and carbohydrates (34 %) was also present. Additionally, they provide essential amino acids such as threonine, lysine and methionine, which are often limited in plant-based foods (61). Nutritional analysis of Swiss chard microgreens revealed a high potassium content (92 g/kg dry weight (DW)) along with significantly low amounts of nitrogen and phosphorus, supporting cardiovascular health (36). Furthermore, bioactive compounds such as chlorophylls, carotenoids, phenolics and anthocyanins were identified using spectrophotometric methods. Low nitrate levels (1061 mg/kg fresh weight (FW)), determined using colorimetric methods, further enhance their suitability for consumption (36). In red amaranth microgreens, antioxidants such as carotenoids and ascorbic acid, along with minerals including potassium, magnesium, iron, sodium, zinc and calcium, were analysed using spectrophotometric, titrimetric and atomic absorption spectroscopy (AAS) methods. The results indicated high nutritional value with elevated antioxidant and mineral content (32). Further studies investigated the influence of light parameters including spectrum, intensity and photoperiod, in the nutritional composition of red amaranth and leafy vegetable amaranth microgreens (37). Exposure to a red and blue light spectrum (70R:30B) at 280 µmol m⁻²s⁻¹ PPFD for 16-hr photoperiod resulted in increased levels of vitamin C, anthocyanins, total carotenoids, chlorophyll a and b and overall antioxidant capacity (37). These findings suggest that optimised light conditions enhance both growth and nutritional quality.

In beet microgreens, key phytochemical components such as total chlorophylls (516.18 mg/100 g DW), carotenoids (114.00 mg/100 g DW), betalains (1264.83 mg/100 g DW), ascorbic acid (1083.30 mg/100 g DW) and polyphenols (908.16 mg gallic acid equivalents (GAE)/100 g DW), reached peak levels on the 15th day of germination (33). Another study reported, red beet microgreens are rich in flavonoids (1625 mg/100 g FW), citric acid (358.83 mg/100 g FW) and essential minerals such as iron (677 µg/100 g FW), zinc (96.74 µg/100 g FW) and manganese (301.05 µg/100 g FW), exhibiting 77.31 % antioxidant activity. However, they also contained relatively high nitrate levels (636.41 mg/kg FW), which may contribute to their nutritional and physiological benefits when consumed within recommended dietary limits (63). Additional studies have shown that beet microgreens contain significant levels of phenolics (13.16 mg GAE g⁻¹ FW) and betalains (0.61 mg g⁻¹ FW), further supporting their antioxidant potential (35). Overall, these findings demonstrate that Amaranthaceae microgreens are nutrient-dense functional foods, providing a concentrated source of phytonutrients compared to their mature counterparts.

Bioactive compounds with health benefits

Several bioactive compounds, including polyphenols, flavonoids and betalains, were found in various microgreens species and contribute to their health-promoting properties. Microgreens have been reported to exhibit antioxidant, anti-inflammatory, antimicrobial, antidiabetic and anticancer activities, as demonstrated through both *in vitro* and *in vivo* studies (52, 65–67). The major bioactivities associated with Amaranthaceae microgreens and their key bioactive compounds are summarised in this section (Table 3).

Table 2. Nutritional composition of selected Amaranthaceae microgreens

Microgreens	Nutrients	Content	References
Quinoa	Tocopherols	65 µg/g	(61)
	β-carotene	738 µg/g	
	Fatty acids-α-linolenic acid	35.1 %	
	Linolenic acid	11.36 %	
	Oleic acid	5.11 %	
Swiss chard	Sucrose	0.091 mg/g	(40)
	Reducing sugar	0.75 mg/g	
	Sugars	2.2 mg/g	
	Phenolics	164 µg/g	
	Chlorophylls	0.771 mg/g	
	Carotenoids	0.122 µg/g	
	Anthocyanins	11.78 µg/g	
Amaranth	Ascorbic acid	0.031 mg/g	(32, 37, 62)
	Chlorophyll a	0.25 mg/g	
	Chlorophyll b	0.20 mg/g	
	Carotenoids	0.023 mg/g	
	Anthocyanins	9 mg/100 g	
Red beets	Potassium	252.70 mg/100 g FW	(34, 63, 64)
	Magnesium	49.28 mg/100 g FW	
	Calcium	67.18 mg/100 g FW	
	Phosphorus	2.57 mg/100 g FW	
	Iron	677 µg/100 g FW	
	Manganese	301.05 µg/100 g FW	
	Zinc	96.74 µg/100 g FW	
	Copper	486.53 µg/100 g FW	
	Glucose	0.580 mg/100 g FW	
	Fructose	0.671 mg/100 g FW	
	Sucrose	0.248 mg/100 g FW	
	Citric Acid	358.83 mg/100 g FW	
	Ascorbic Acid	32.72 mg/100 g FW	
	Betaxanthins	432.7 mg/100 g	
	Betacyanin	226.7 mg/100 g	
Spinach	Polyphenols	313.8 mg/100 g	(45, 57)
	Chlorophylls	44 µg/g	
	Lutein	54.2 µg/g	
	β-carotene	44 µg/g	
	Phenols	632.3 µg/g	
	Ascorbic acid	130.5 µg/g	
	α-tocopherol	14.2 ± 3.3 mg/100 g FW	
	γ-tocopherol	5.1 ± 0.8 mg/100 g FW	

FW: fresh weight.

Table 3. Functional properties of Amaranthaceae microgreens and associated bioactive compounds

Functional property	Microgreens	Key bioactive compounds	References
Anti-microbial	<i>S. oleracea</i>	Phenolic compounds	(68)
Antioxidant	<i>Amaranthus hypochondriacus</i> L, <i>B. vulgaris</i> , <i>C. quinoa</i> , <i>S. oleracea</i>	Phenolics, Betalains, Anthocyanin and Carotenoids	(33, 45, 48, 57, 61)
Cardiovascular health	<i>B. vulgaris</i> , <i>B. vulgaris</i> subsp. <i>cicla</i>	Potassium, Flavonoids and Phenolics	(36, 64, 69)
Anti-inflammatory	<i>B. vulgaris</i>	Betalains (Betacyanins and Betaxanthins)	(64)
Gastrointestinal health	<i>B. vulgaris</i> , <i>S. oleracea</i>	Polyphenols	(58, 68)

Quinoa microgreens are rich in polyphenols and flavonoids, exhibiting potent antioxidant properties, which help reduce oxidative stress and may lower the risk of chronic conditions such as cardiovascular diseases and metabolic syndrome (61). Similarly, red amaranth microgreens contain high levels of carotenoids (49.96 ± 0.87 mg/100 g FW) and ascorbic acid (90.3 ± 1.2 mg/100 g FW), supporting immune function and protecting against oxidative damage (32). In addition, red amaranth microgreens possess high polyphenol content (769.8 mg gallic acid equivalent/100 g DW) and strong antioxidant capacity (3154.2 mg trolox equivalent/100 g DW). These properties are attributed to the presence of flavonoids and

resveratrol, which also contribute to significant tyrosinase inhibitory activity (17969.3 mg kojic acid equivalents/100 g DW), suggesting neuroprotective effects and enhancing their value as functional foods (62).

Red beet microgreens are particularly rich in betalains (249.1 mg/100 g DW), which contribute to their antioxidant and anti-inflammatory properties. Their phenolic content (432.7 mg/100 g DW) has been reported to increase during storage, with approximately a 1.3-fold rise by the 10th day, potentially enhancing their positive effect on cardiovascular health (70). Due to their rich phytochemical and mineral composition, red beet microgreens

provide multiple health benefits. Their high flavonoid content (1625 mg/100 g FW) supports anti-inflammatory and cardiovascular functions, while essential minerals such as iron and zinc contribute to immune health. Additionally, citric acid (358.83 mg/100 g FW) may aid digestive health (70). Other studies have reported higher levels of betalain (1264.83 mg/100 g), along with polyphenols and carotenoids, which collectively contribute to antioxidant, antidiabetic and potential cancer-preventive effects, and highlighting beet microgreens as highly functional foods (33). The antimicrobial properties of red spinach microgreens have also been evaluated using ethanolic extracts against selected pathogenic bacteria. Using micro-dilution technique to determine minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC), the extracts showed significant inhibitory effects, particularly against *Bacillus subtilis* and *Escherichia coli*. These findings suggest their potential role in enhancing food safety and supporting human health (68). Overall, these findings highlight the functional significance of Amaranthaceae microgreens as valuable dietary components that support health promotion and disease prevention through their rich bioactive composition.

Conclusion

Amaranthaceae microgreens are a promising, nutrient-dense functional food category with the potential to enhance both human health and support sustainable agriculture. They are rich in essential nutrients and bioactive compounds with antimicrobial, anti-inflammatory and antioxidant properties. Their yield and nutritional value are greatly improved through the use of advanced cultivation techniques, such as hydroponics and controlled-environment systems. By optimising factors such as photoperiod, light quality and nutrient delivery, these systems enhance the accumulation of bioactive compounds while minimising resource use. Therefore, microgreens offer a sustainable and feasible alternative to mature leafy vegetables, particularly in the context of resource-constrained agricultural and urban farming. Despite their potential, research on Amaranthaceae microgreens remains limited. Future studies should focus on expanding species diversity, optimising cultivation and post-harvest practices and evaluating long-term health benefits. Investigating consumer acceptance and market potential is also essential, particularly for underutilised species within the family. The inclusion of Amaranthaceae microgreens in daily diets may play a significant role in improving public health, enhancing food security and reducing environmental pressures associated with conventional agriculture.

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Authors' contributions

PB conducted the conceptualisation and drafting of the manuscript. SSM and SWK contributed review and editing. All authors read and approved the final manuscript.

Compliance with ethical standards

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